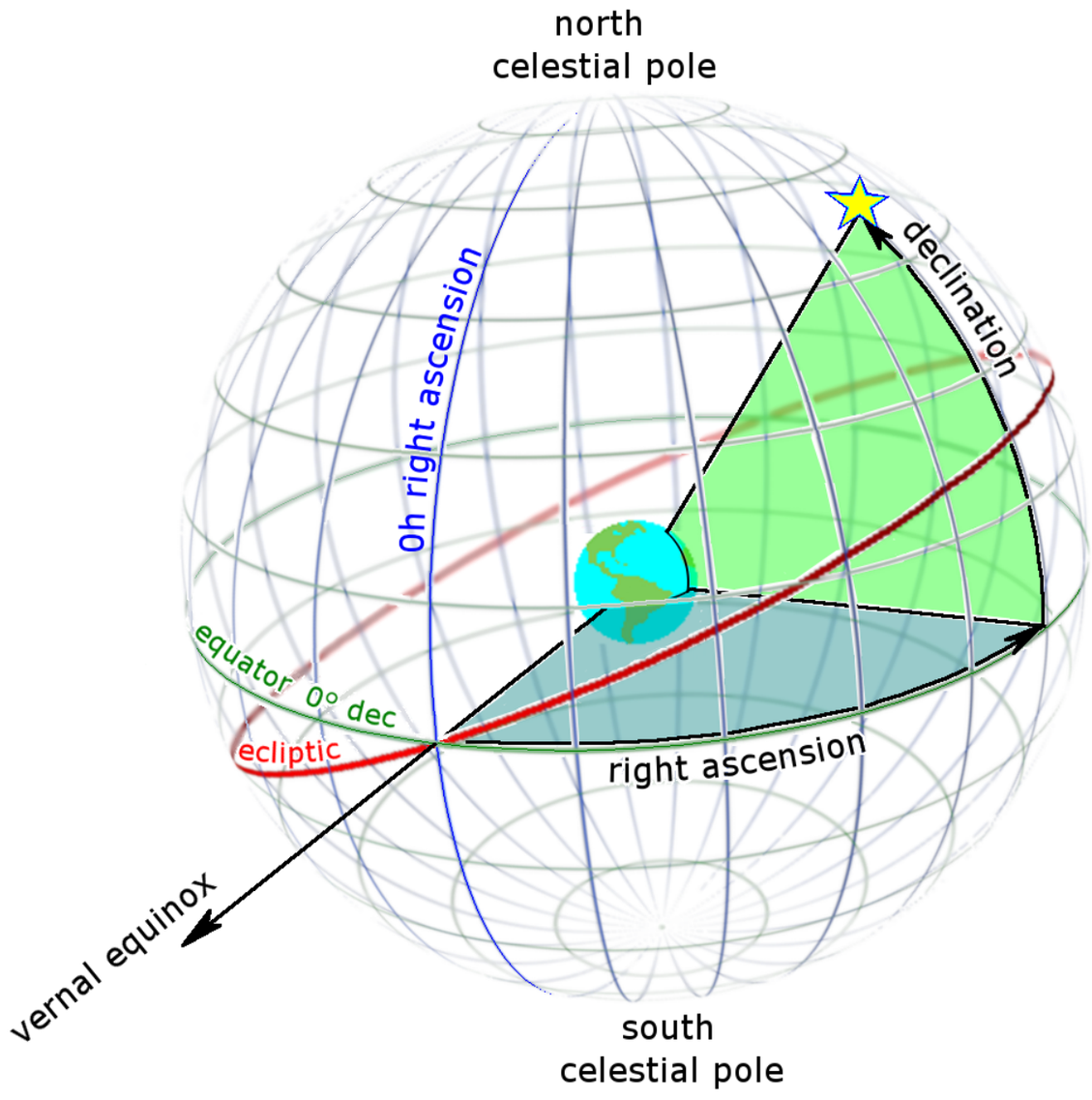
Spherical Coordinates

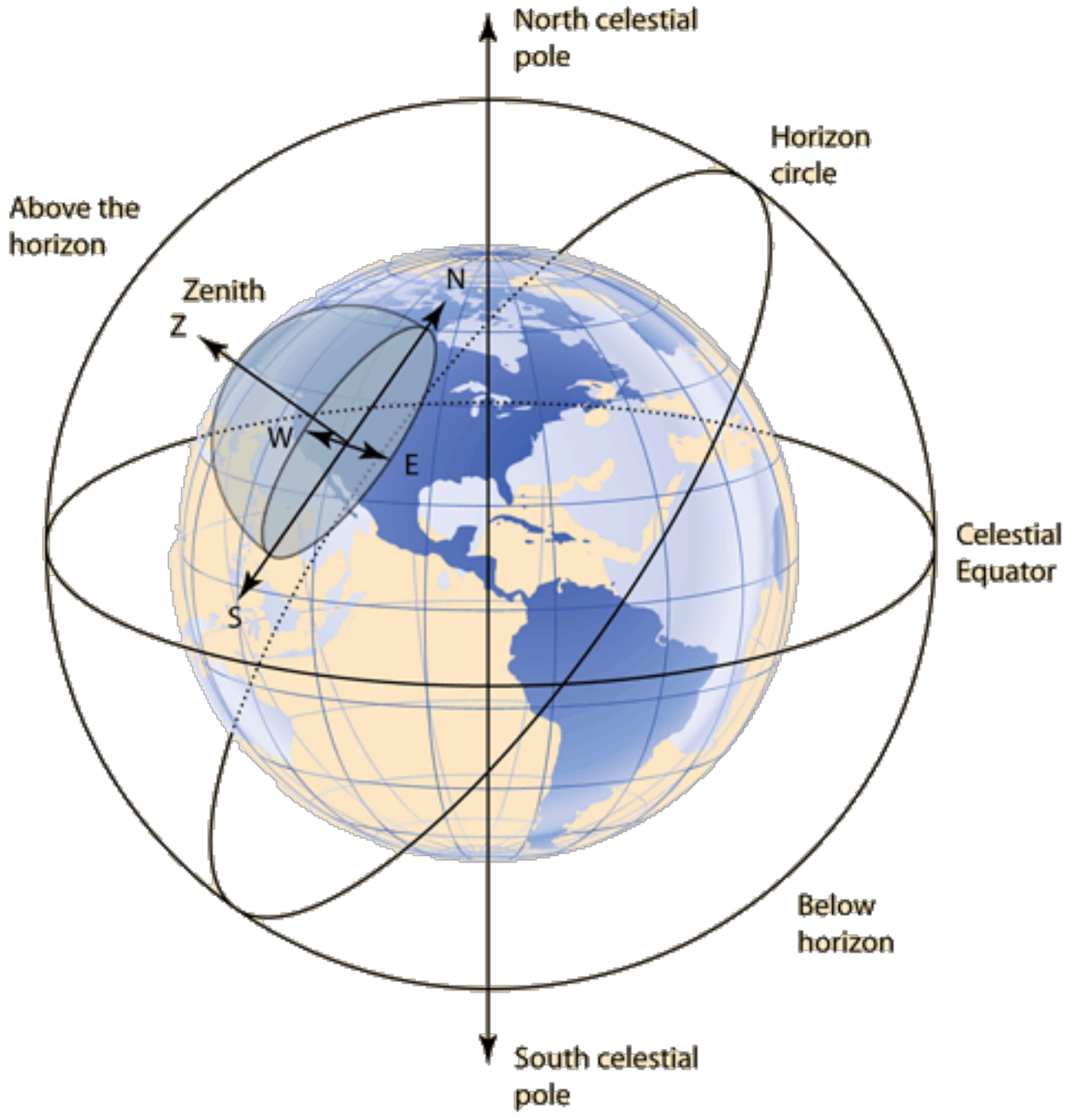
Spherical Coordinates

- Equatorial:
  - Right Ascension (RA) and Declination are based on the Earth's equator
  - Because of precession of the equinox, the location of RA=0hr changes (on a 26,000 year cycle). So coordinates are referred to a specific epoch, e.g. J2000.
  - *Beware* coordinates in *1950*
  - **Gotcha:** sexagesimal to decimal conversion



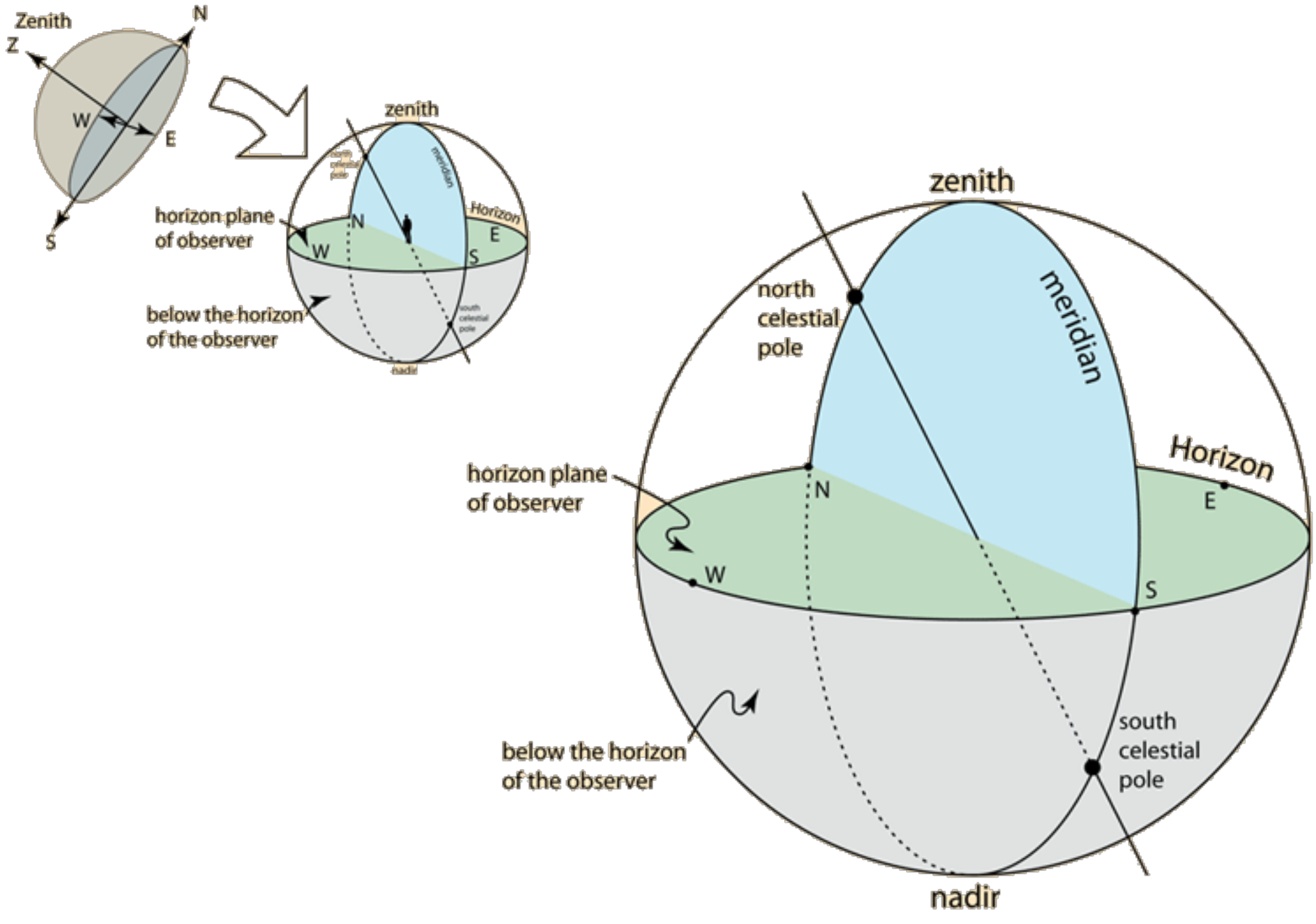
By Tfr000 (talk) 15:34, 15 June 2012 (UTC) - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=19907447

What's up in the sky?



- Observations depend on altitude and azimuth: transformation of coordinate systems

What's up in the sky?

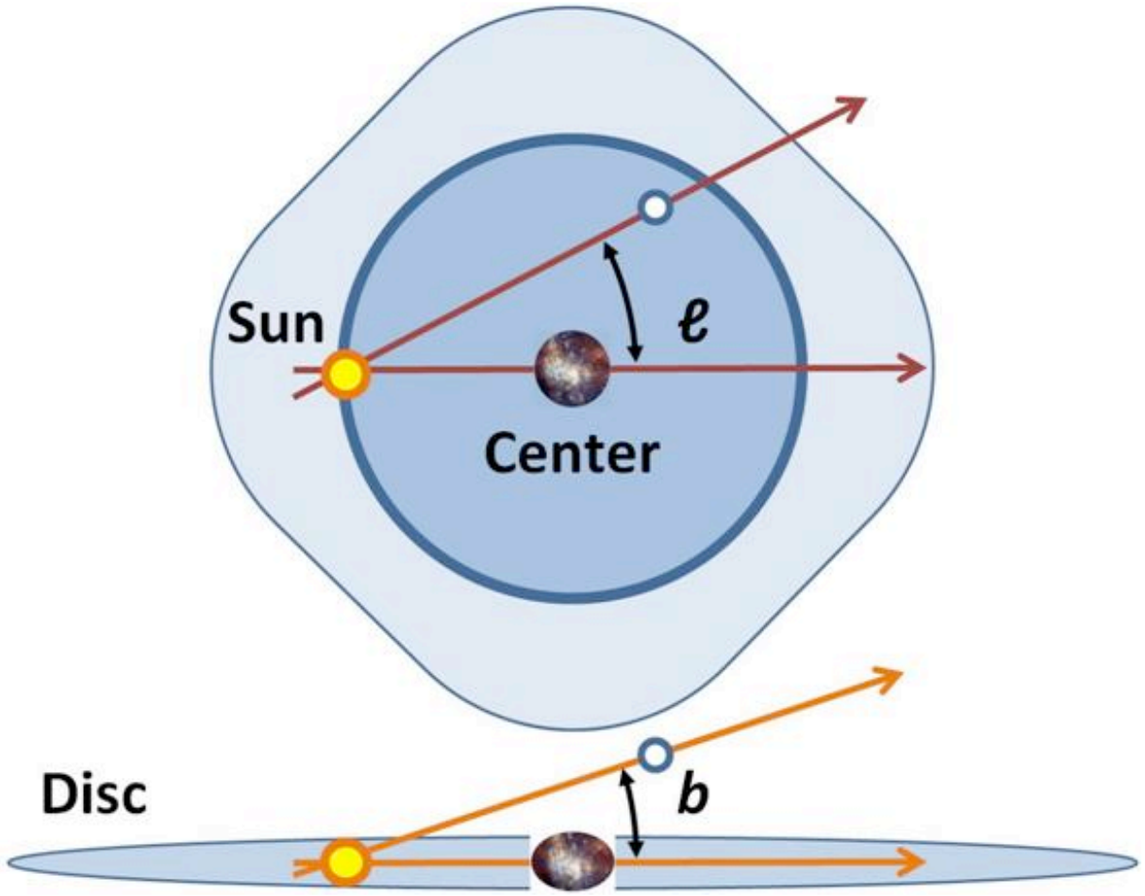


Equatorial

- Observing: if an object with RA=10h is rising in the East, then an object with RA=11h will rise one hour later.
- **Gotcha**: As seen on the sky longitude coordinates go "*backwards*":
  - RA increases to the *left* if North is up.
  - (Same for Galactic coords)

Spherical Coordinates

- Galactic:
  - longitude: l = 0 towards galactic centre, 90 in direction of rotation
  - latitude with respect to plane of Galaxy



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Spherical Trig

- Angular separation θ between 2 points with longitude λ and latitude ϕ

- Haversine formula:

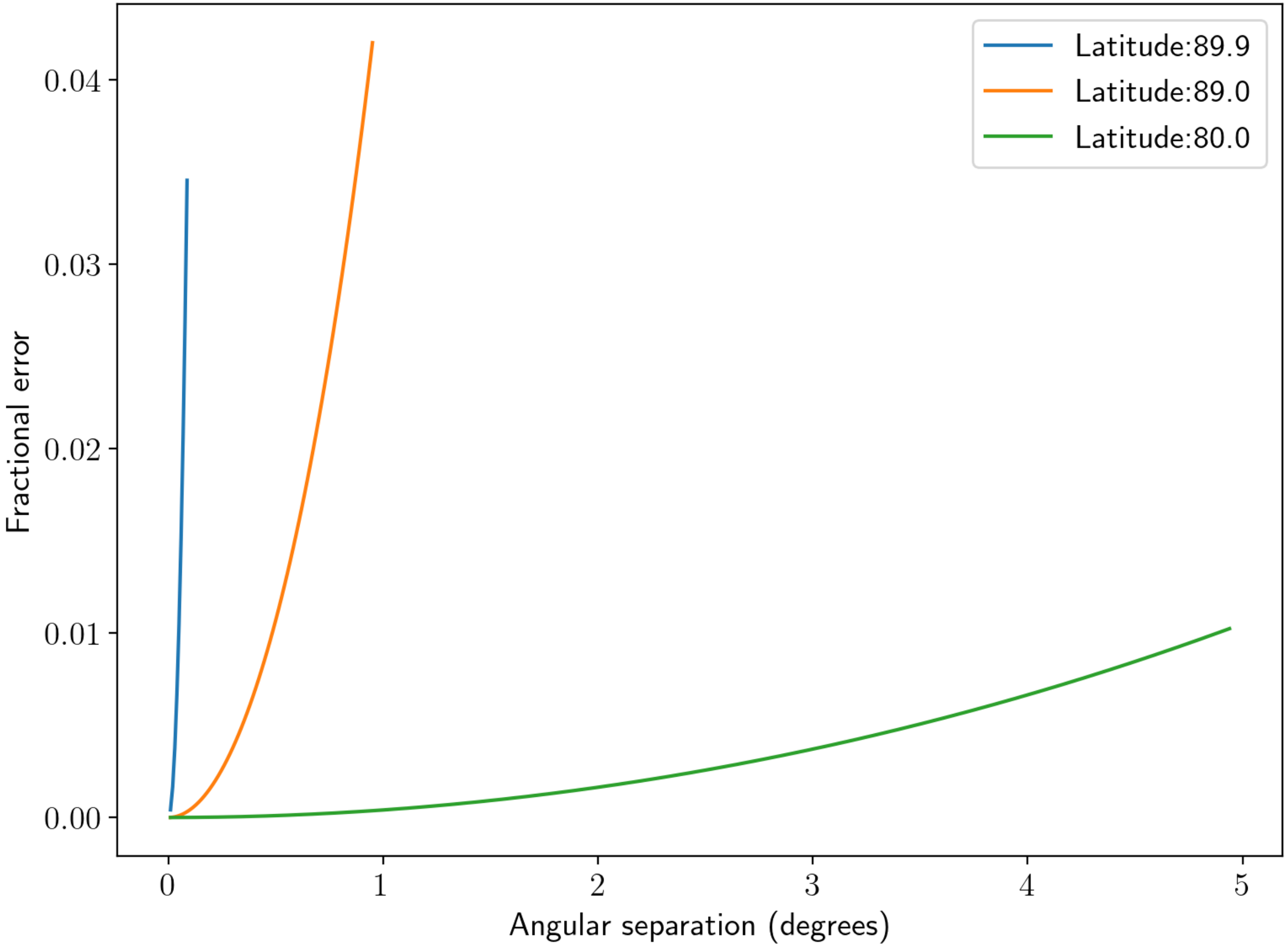
$$\theta = 2\arcsin\left(\sqrt{\sin^2\left(\frac{\varphi_2 - \varphi_1}{2}\right) + \cos(\varphi_1)\cos(\varphi_2)\sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right)$$

- If close together and far from pole can approximate with

$$x = -\cos\left(\frac{\phi_1 + \phi_2}{2}\right)(\lambda_2 - \lambda_1)$$

$$y = \phi_2 - \phi_1$$

$$\theta = \sqrt{x^2 + y^2}$$


Latitude:89.9
Latitude:89.0
Latitude:80.0
0.04
0.03
0.02
0.01
0.00
Fractional error
0
1
2
3
4
5
Angular separation (degrees)

Cartesian to Spherical

$$x = \cos(l)\cos(b) \qquad b = \arcsin(z)$$

$$y = \sin(l)\cos(b) \qquad l = \arctan(y/x)$$

$$z = \sin(b)$$

- `l= atan(y/x)`?
  - Gotcha!
    - limited to -pi/2 to pi
    - Use `atan2(y,x)`

- All trig functions use **radians**

Conversions

- Can be converted according to first principles (see text 4.1)
- Better to use:
    - `astropy.coordinates`
    - NASA Extragalactic Database (NED) calculators

Projections

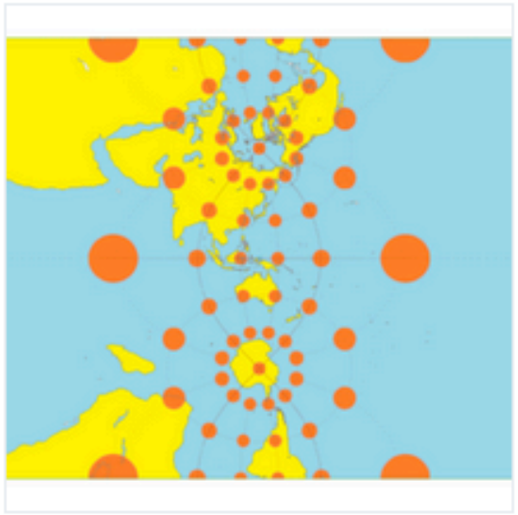
- The spherical sky cannot be displayed on a flat 2D surface (such as a CCD) without distortion

- One example is the gnomonic or tangent plane projection:

- Draw rays from the centre through the sphere to a plane that is tangent to the surface.

- The mapping between the coordinates on the surface of the sphere and the x,y points in the plane is the gnomonic projection.

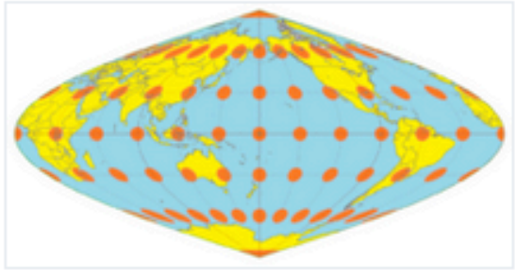
Map projections



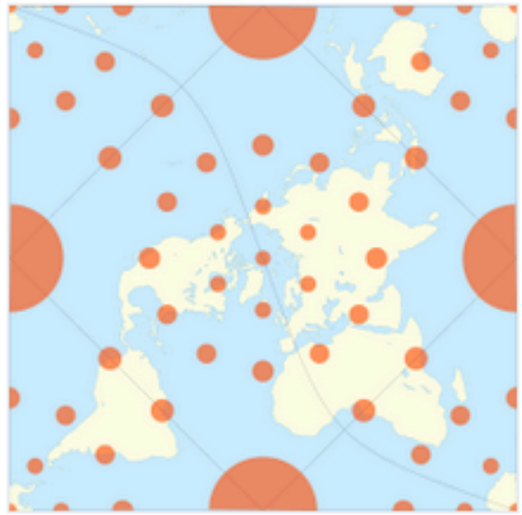
The transverse Mercator projection with Tissot's indicatrices



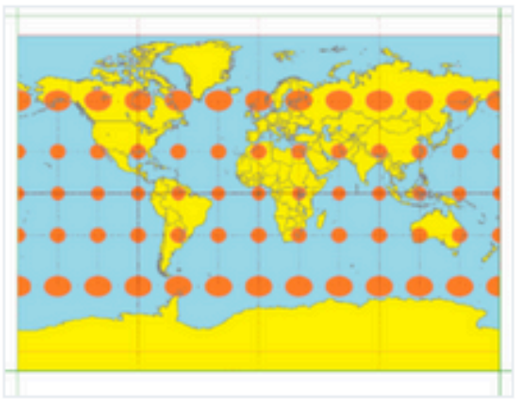
The stereographic projection with Tissot's indicatrices



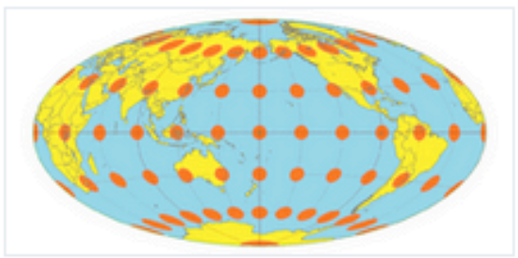
The sinusoidal projection with Tissot's indicatrices



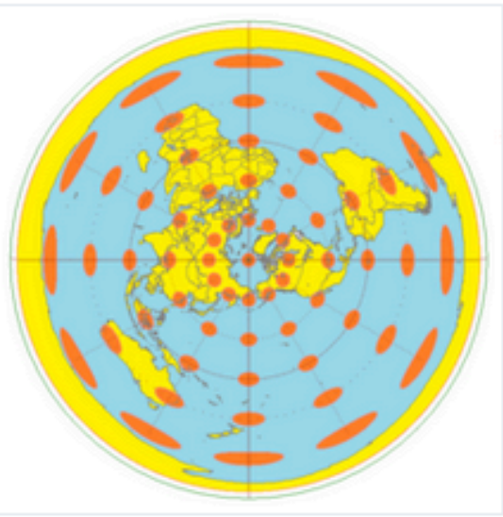
The Peirce quincuncial projection with Tissot's indicatrices



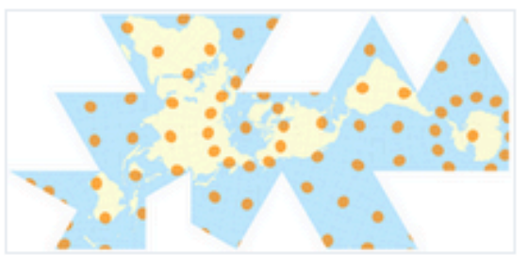
The Miller cylindrical projection with Tissot's indicatrices



The Hammer projection with Tissot's indicatrices



The azimuthal equidistant projection with Tissot's indicatrices



The Fuller projection with Tissot's indicatrices

Calibrating the WCS

- In astronomy, the conversion between x,y in pixels to coordinates on the sky (**astrometry**) is usually handled by the World Coordinate System standard.

- To calibrate the WCS, we need astrometric reference stars. These days, these come from the GAIA mission.

- There are several packages which do this more or less automatically (e.g. SCAMP)

- You can access the WCS in an image via `astropy.wcs`

Fast operations on a sphere

- If you have large data sets over a wide angle, the approximations may not be sufficient.

- One library (widely used in CMB) is HEALPix, which has a Python interface healpy